

EE-102: Electric Circuit Analysis

LECTURE NO 10: *R, L and C Elements and the Impedance Concept*

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1. Introduction

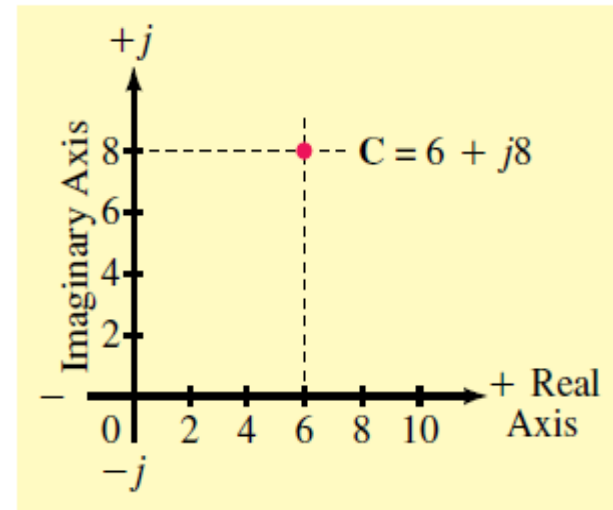
- To analyze ac circuits in the time domain using voltages and currents expressed as functions of time is not very practical.
- It is *more practical* to:
 - *Express voltages and currents as phasors.*
 - *Circuit elements as impedances.*
 - *Represent them using complex numbers.*
- With this, ac circuit analysis is handled much like dc circuit analysis, and all basic relationships and theorems (i.e., KVL, KCL, Mesh Analysis etc.) apply.

2. Complex Number Review (1/10)

- A complex number has the form:
 - $C = a + jb$, where $j = \sqrt{-1}$ (mathematics uses i to represent imaginary numbers)
 - a is the real part
 - jb is the imaginary part

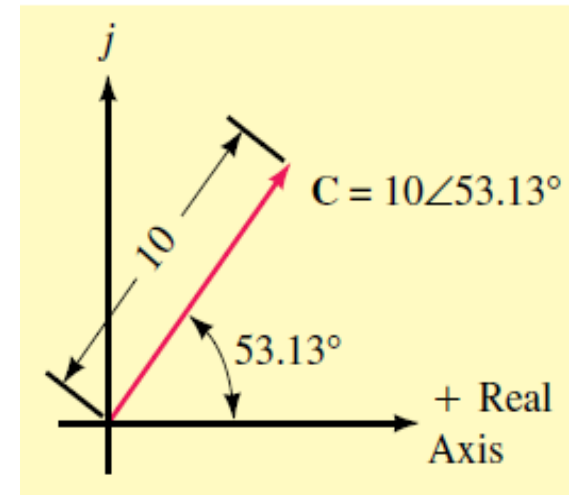
2. Complex Number Review (2/10)

- Complex number
 - May be represented geometrically, either in *rectangular form* or in polar form on a two-dimensional plane called the *complex plane*.
 - The complex number $C = 6 + j8$ is represented geometrically in *rectangular form*.



2. Complex Number Review (3/10)

- May also be represented in *polar form* by magnitude and angle.
- The complex number $\mathbf{C} = 10\angle 53.13^\circ$ has a magnitude of 10 and angle 53.13°
- The complex number $\mathbf{C} = 10\angle 53.13^\circ$ is represented geometrically in *Polar form*.



2. Complex Number Review (4/10)

Conversion between Rectangular and Polar forms

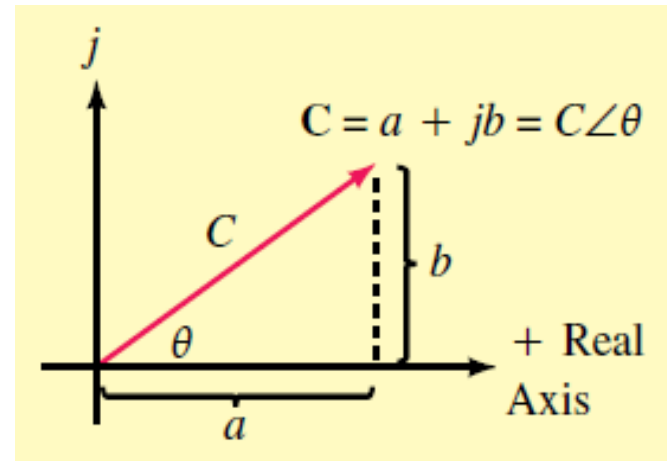
- $C = a + jb$ (rectangular form)
- $C = C \angle \theta$ (polar form)

$$a = C \cos \theta$$

$$b = C \sin \theta$$

$$C = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1} \frac{b}{a}$$



2. Complex Number Review (5/10)

- $j^0 = 1$
- $j^1 = j$
- $j^2 = -1$
- $j^3 = j^2 j = -j$
- $j^4 = j^2 j^2 = (-1)(-1) = 1$
- $(-j)j = -j^2 = 1$
- $1/j = 1/j \times j/j = j/j^2 = j/-1 = -j$

2. Complex Number Review (6/10)

Addition and Subtraction of Complex Numbers

- Addition and subtraction of complex numbers can be performed *analytically* and *graphically*.
- *Analytically*, add real and imaginary parts separately and same procedure for subtraction.
- *Graphically* add vectorially , for subtraction change the sign of the subtrahend and then add.

2. Complex Number Review (7/10)

EXAMPLE 1

Given $\mathbf{A} = 2 + j1$ and $\mathbf{B} = 1 + j3$, determine their sum and difference analytically and graphically.

Solution

$$\mathbf{A} + \mathbf{B} = (2 + j1) + (1 + j3) = (2 + 1) + j(1 + 3) = 3 + j4.$$

$$\mathbf{A} - \mathbf{B} = (2 + j1) - (1 + j3) = (2 - 1) + j(1 - 3) = 1 - j2.$$

2. Complex Number Review (8/10)

Graphical addition and subtraction are shown in Figure 1.

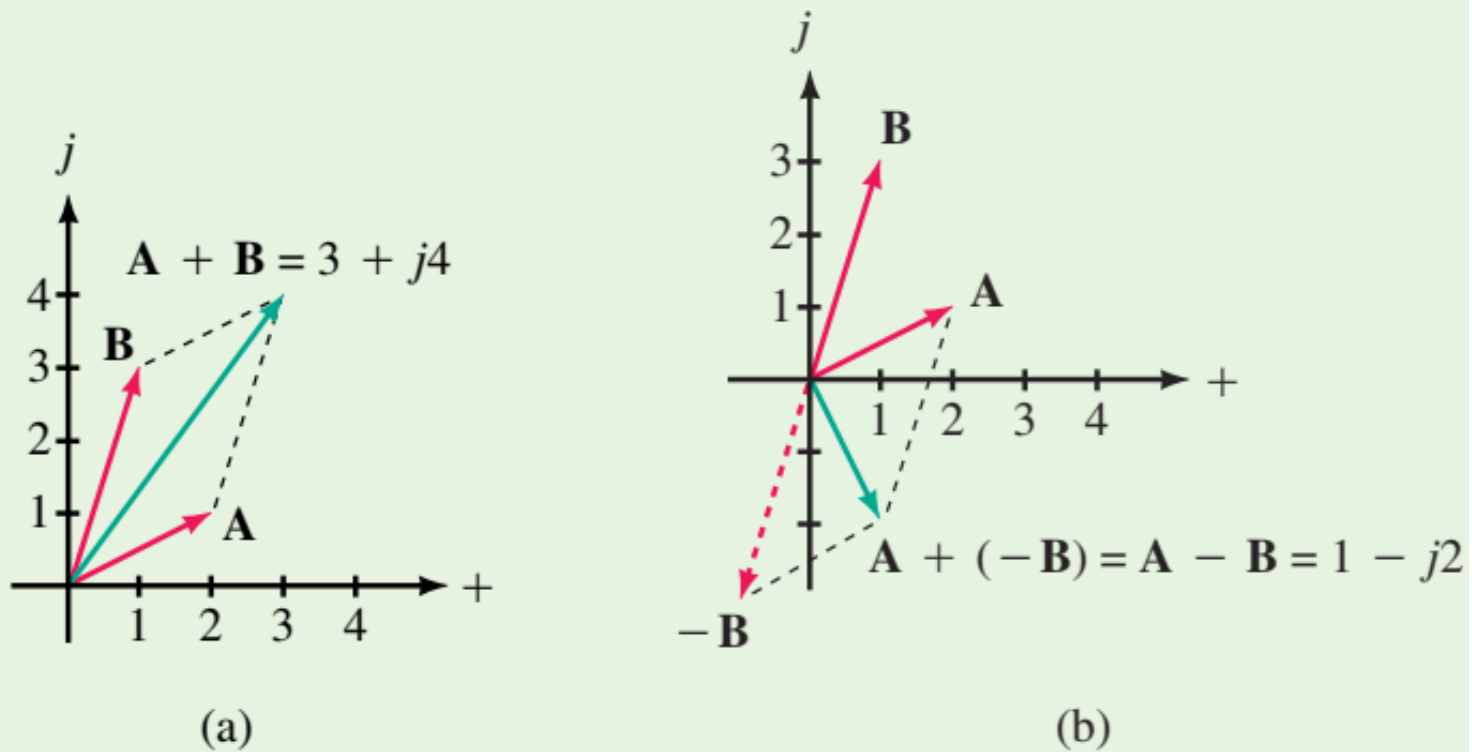


Figure: 1

2. Complex Number Review (9/10)

Multiplication and Division of Complex Numbers

- Multiplication and division operations are usually performed in polar form
- For *Multiplication*

$$\mathbf{A \cdot B = AB/\theta_A + \theta_B}$$

- For *Division*

$$\mathbf{A/B = A/B/\theta_A - \theta_B}$$

- The *reciprocal* of a complex number $C = C\angle\theta$ is:

$$\frac{1}{C\angle\theta} = \frac{1}{C}\angle-\theta$$

- *Conjugate* of a complex number is

$$\mathbf{C = C\angle\theta = a + jb \text{ is } C^* = C\angle-\theta = a - jb}$$

2. Complex Number Review (10/10)

EXAMPLE 2

Given $\mathbf{A} = 3\angle 35^\circ$ and $\mathbf{B} = 2\angle -20^\circ$, determine the product $\mathbf{A} \cdot \mathbf{B}$ and the quotient $\mathbf{A}/\mathbf{B}$.

Solution

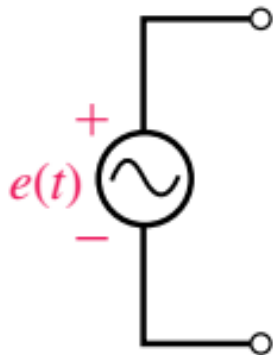
$$\mathbf{A} \cdot \mathbf{B} = (3\angle 35^\circ)(2\angle -20^\circ) = (3)(2)\angle 35^\circ - 20^\circ = 6\angle 15^\circ$$

$$\frac{\mathbf{A}}{\mathbf{B}} = \frac{(3\angle 35^\circ)}{(2\angle -20^\circ)} = \frac{3}{2}\angle 35^\circ - (-20^\circ) = 1.5\angle 55^\circ$$

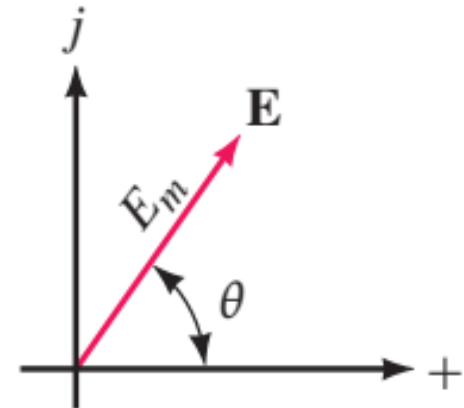
3. Voltages and Currents as Complex Numbers (1/4)

- AC voltages and currents can be represented as phasors.
- Phasors have magnitude and angle
 - Can be viewed as complex numbers

$$\mathbf{E} = E_m \angle \theta$$



$$(a) \ e(t) = E_m \sin(\omega t + \theta)$$



$$(b) \ \mathbf{E} = E_m \angle \theta$$

3.Voltages and Currents as Complex Numbers (2/4)

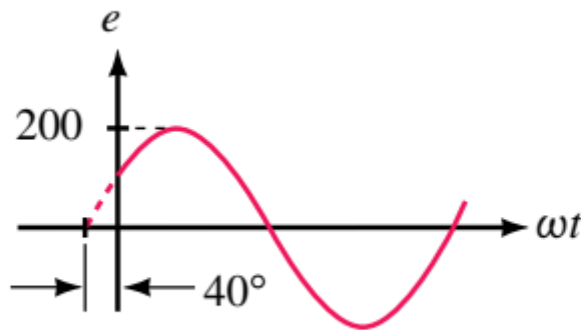
- A voltage given as $e(t) = 200 \sin (\omega t + 40^\circ)$
 - Can be represented by its phasor equivalent, $\mathbf{E} = 200\angle 40^\circ$
- RMS value is used in phasor form so that power calculations are correct
- $200\angle 40^\circ =$ voltage of 200 volts rms or effective value at an angle of 40°
- $200\angle 40^\circ = V_{eff}\angle 40^\circ = 0.707V_m\angle 40^\circ = 0.707(200)\angle 40^\circ$

Above voltage would be written as $141.4\angle 40^\circ$

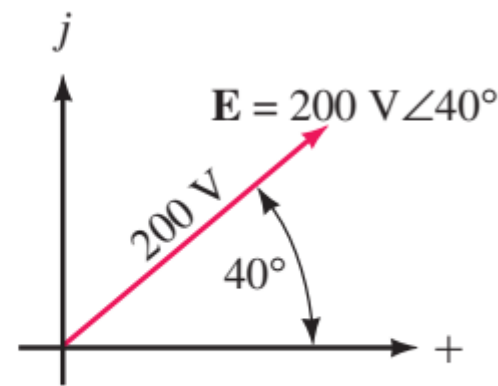
$$E_{\text{rms}} = \frac{1}{\sqrt{2}} E_m = 0.707 E_m$$



(a) $e = 200 \sin (\omega t + 40^\circ)$ V



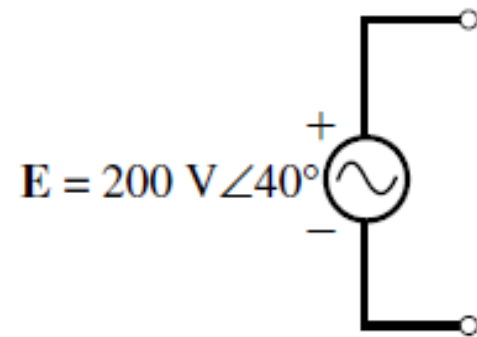
(b) Waveform



(c) Phasor equivalent

3.Voltages and Currents as Complex Numbers (3/4)

- We can represent a (time-varying) source by its phasor equivalent from the start.
- Quantities expressed as time functions are said to be in the *time domain*, while quantities expressed as phasors are said to be in the *phasor (or frequency) domain*.
- Phasor representation contains information we need except for angular velocity.



Transformed source

Figure: 3

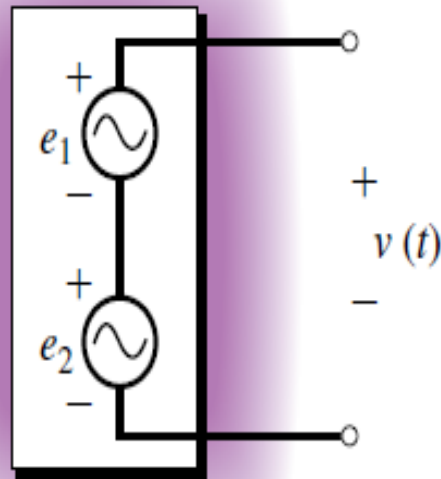
3. Voltages and Currents as Complex Numbers (4/4)

- By doing this, we have transformed from the time domain to the phasor domain (frequency domain).
- KVL and KCL
 - Apply in both time domain and phasor domain
 - $e = v_1 + v_2$ (Time domain)
 - $E = V_1 + V_2$ (Phasor domain)
 - Similarly for currents.

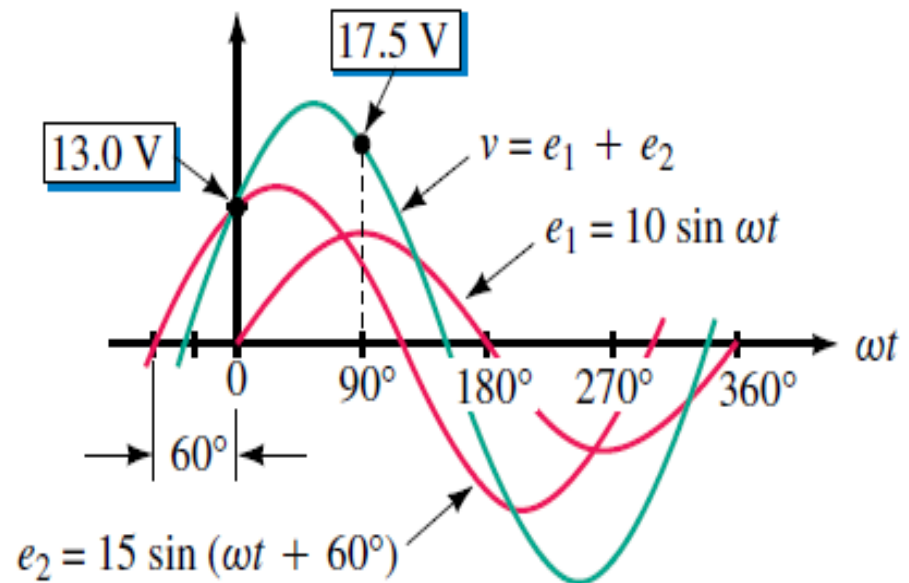
4. Summing AC Voltages and Currents (1/5)

- Sinusoidal quantities must sometimes be added or subtracted.
- For example the sum of e_1 and e_2 can be found by adding two voltage waveforms point by point to get the resultant voltage waveform.
- But adding or subtracting waveforms in time domain is very tedious and provides no analytical expression for resulting voltage.

4. Summing AC Voltages and Currents continue (2/5)



(a) $e_1 = 10 \sin \omega t$
 $e_2 = 15 \sin (\omega t + 60^\circ)$



(b) Waveforms

Figure: 4

4. Summing AC Voltages and Currents continue (3/5)

- A better way is to transform the sources.
 - Replace voltages or currents with their phasor equivalents
 - Add them as complex numbers
- Once waveforms are added
 - Corresponding time equation of resultant waveform can be determined

4. Summing AC Voltages and Currents continue (4/5)

EXAMPLE 4

Given $e_1 = 10 \sin \omega t$ V and $e_2 = 15 \sin(\omega t + 60^\circ)$ V , determine v and sketch it.

Solution

$$e_1 = 10 \sin \omega t \text{ V. Thus, } \mathbf{E}_1 = 10 \text{ V} \angle 0^\circ.$$

$$e_2 = 15 \sin(\omega t + 60^\circ) \text{ V. Thus, } \mathbf{E}_2 = 15 \text{ V} \angle 60^\circ.$$

$$\mathbf{V} = \mathbf{E}_1 + \mathbf{E}_2 = 10 \angle 0^\circ + 15 \angle 60^\circ = (10 + j0) + (7.5 + j13) = (17.5 + j13)$$

$$= 21.8 \text{ V} \angle 36.6^\circ \text{ Thus, } v = 21.8 \sin(\omega t + 36.6^\circ) \text{ V.}$$

Waveforms are shown in (c) next slide.

4. Summing AC Voltages and Currents continue (5/5)

Transformed sources are shown in Figure 5(a) and phasors in (b).

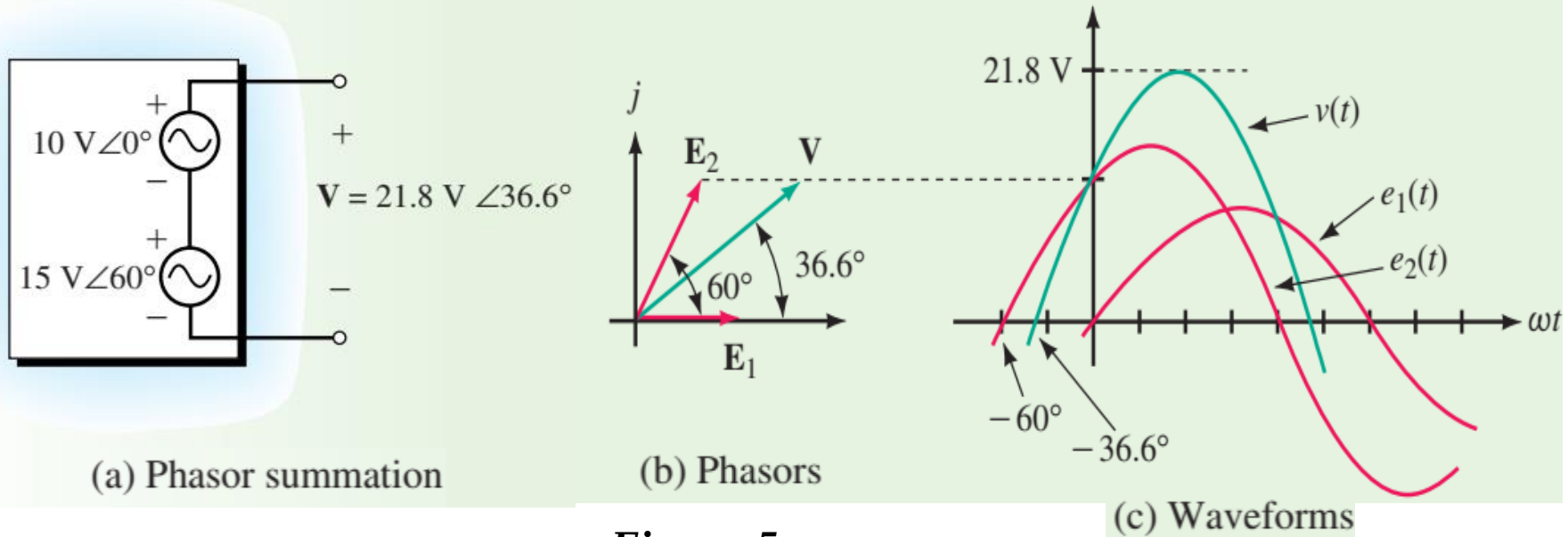


Figure: 5

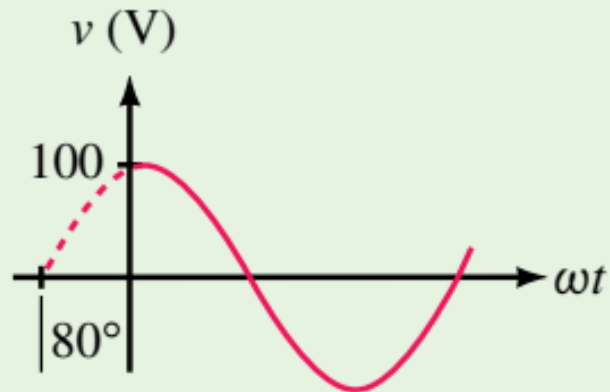
Refer to slide no 7 (for equations) in order to solve this example.

Important Notes

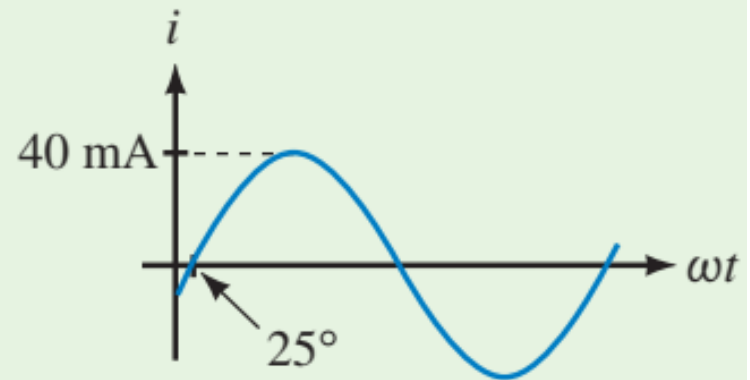
- Until now, we have used peak values such as V_m and I_m to represent the magnitudes of phasor voltages and currents.
- It is more common to write them as RMS values
- To add or subtract sinusoidal voltages or currents
 - Convert to phasor form, add or subtract, then convert back to sinusoidal form
- Quantities expressed as phasors
 - Are in phasor domain or frequency domain

EXAMPLE 5

Express the voltage and current of Figure 6 in both the time and the phasor domains. (Remember to use rms values in the phasor domain.)



(a)



(b)

Figure: 6

EXAMPLE 5

Solution

a. Time domain: $v = 100 \sin(\omega t + 80^\circ)$ volts.

Phasor domain: $\mathbf{V} = (0.707)(100 \text{ V} \angle 80^\circ) = 70.7 \text{ V} \angle 80^\circ$ (rms).

b. Time domain: $i = 40 \sin(\omega t - 25^\circ)$ mA.

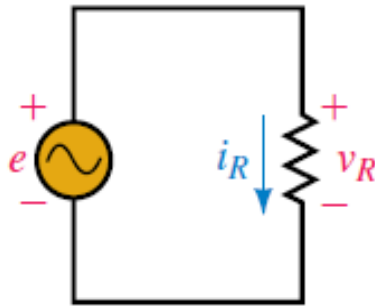
Phasor domain: $\mathbf{I} = (0.707)(40 \text{ mA} \angle -25^\circ) = 28.3 \text{ mA} \angle -25^\circ$ (rms).

4. R , L , and C Circuits with Sinusoidal Excitation

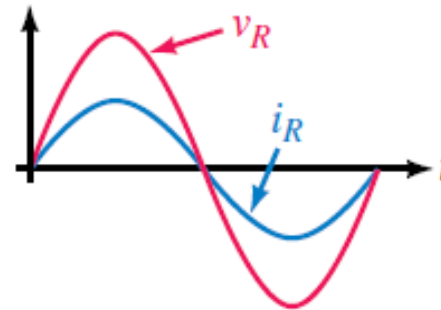
- R , L , and C circuit elements
 - *Have different electrical properties*
 - *Differences result in different voltage-current relationships*
- When a circuit is connected to a sinusoidal source
 - All currents and voltages will be sinusoidal
- *These sine waves will have the same frequency (and therefore same time period) as the source*
 - *Only difference is their magnitudes and angles*

4.1 Resistance and Sinusoidal AC

- In a purely resistive circuit
 - Ohm's Law applies, thus current is proportional to the voltage.
 - Current variations follow voltage variations.
 - *For a purely resistive circuit, current and voltage are in phase.*



(a) Source voltage is a sine wave.
Therefore, v_R is a sine wave.



(b) $i_R = v_R/R$. Therefore
 i_R is a sine wave.

Resistance and Sinusoidal AC continue...

- The relationship illustrated in figure (previous slide) may be stated mathematically as:

$$i_R = \frac{v_R}{R} = \frac{V_m \sin \omega t}{R} = \frac{V_m}{R} \sin \omega t = I_m \sin \omega t$$

where

$$I_m = V_m/R$$

$$V_m = I_m R$$

- The in-phase relationship is true regardless of reference.
 - If $v_R = V_m \sin(\omega t + \theta)$, then $i_R = I_m \sin(\omega t + \theta)$

EXAMPLE 6

For the circuit of Figure 7, if $R = 5\ \Omega$ and $i_R = 12 \sin(\omega t - 18^\circ)\text{ A}$, determine v_R .

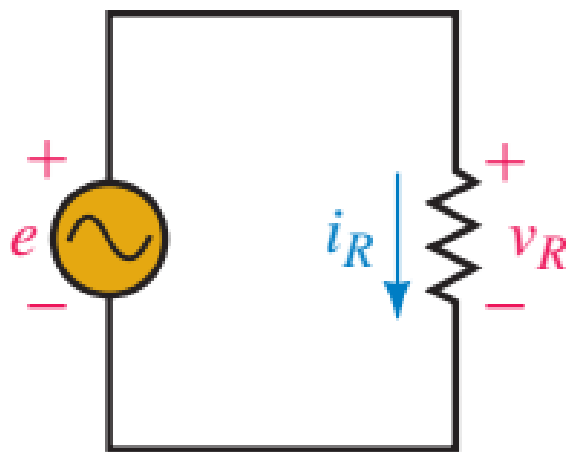


Figure: 7

EXAMPLE 6

Solution $v_R = Ri_R = 5 \times 12 \sin(\omega t - 18^\circ) = 60 \sin(\omega t - 18^\circ)$ V. The waveforms are shown in Figure 8.

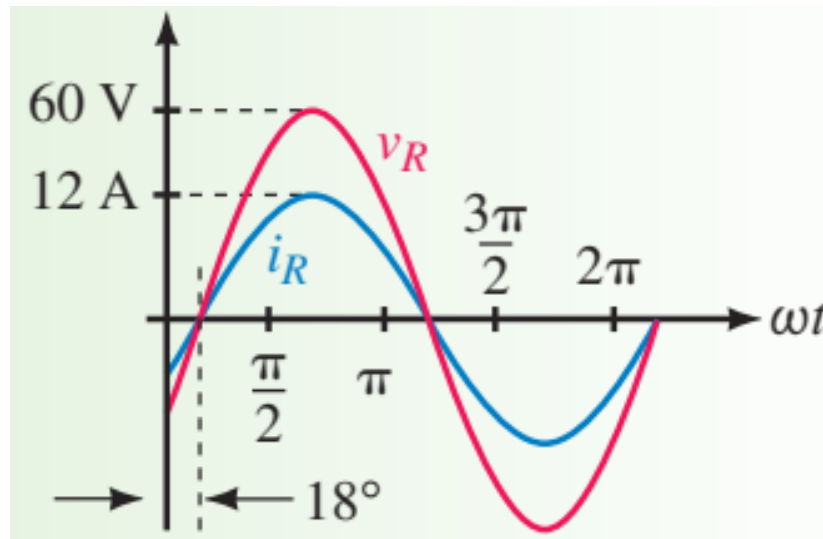
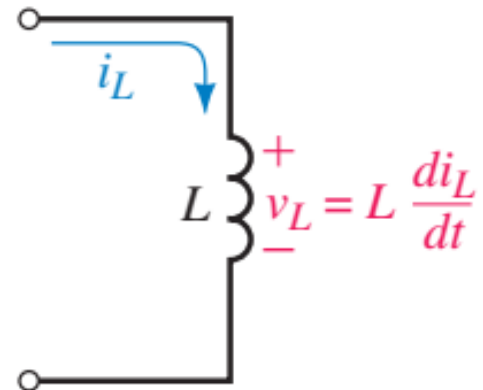


Figure: 8

4.2 Inductance and Sinusoidal AC

- *Phase lag in an Inductive Circuit*
- For an ideal inductor
 - Voltage(v_L) is proportional to rate of change of current
- Voltage is greatest when the rate of change of the current is greatest
 - Voltage and current are not in phase



Inductance and Sinusoidal AC continue...

- For a sine wave of current, you get when you differentiate

$$v_L = L \frac{di_L}{dt} = L \frac{d}{dt}(I_m \sin \omega t) = \omega L I_m \cos \omega t = V_m \cos \omega t$$

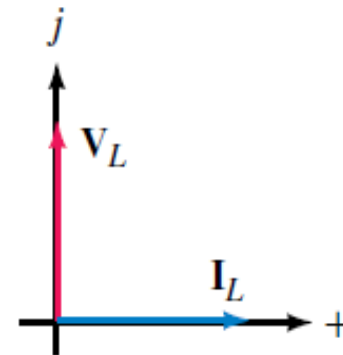
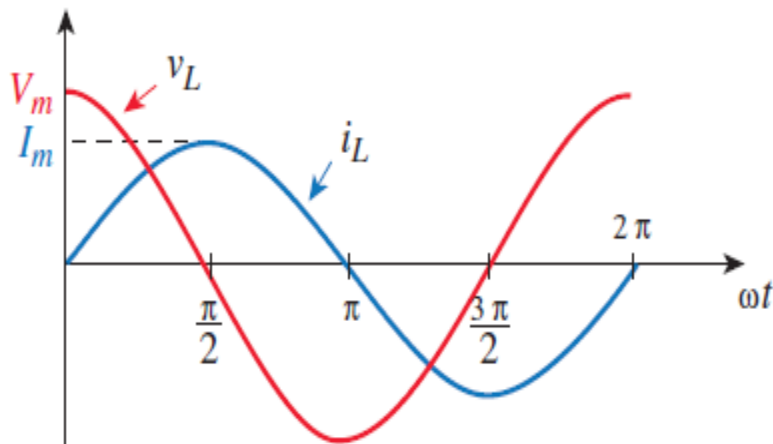
- Using trigonometric identity $\cos \omega t = \sin(\omega t + 90^\circ)$

$$v_L = V_m \sin(\omega t + 90^\circ)$$

$$V_m = \omega L I_m$$

Inductance and Sinusoidal AC continue...

- For a pure inductive circuit, current lags voltage by 90°
- Alternatively, voltage leads current by 90° .



4.2.1 Inductive Reactance

- $V_m = \omega L I_m$
- Using above equation the ratio V_m to I_m is:

$$\frac{V_m}{I_m} = \omega L$$

- This ratio is defined as *inductive reactance* (X_L).
- Ratio of volts to amps is ohms

$$X_L = \frac{V_m}{I_m} \text{ } (\Omega) \longrightarrow X_L = \omega L \text{ } (\Omega)$$

- Current and voltage amplitudes are related by:

$$I_m = \frac{V_m}{X_L} \quad \& \quad V_m = I_m X_L$$

EXAMPLE 7

The voltage across a 0.2-H inductance is $v_L = 100 \sin (400t + 70^\circ)$ V. Determine i_L and sketch it.

Solution $\omega = 400$ rad/s. Therefore, $X_L = \omega L = (400)(0.2) = 80 \Omega$.

$$I_m = \frac{V_m}{X_L} = \frac{100 \text{ V}}{80 \Omega} = 1.25 \text{ A}$$

EXAMPLE 7

The current lags the voltage by 90° . Therefore $i_L = 1.25 \sin(400t - 20^\circ)$ A as indicated in Figure 9

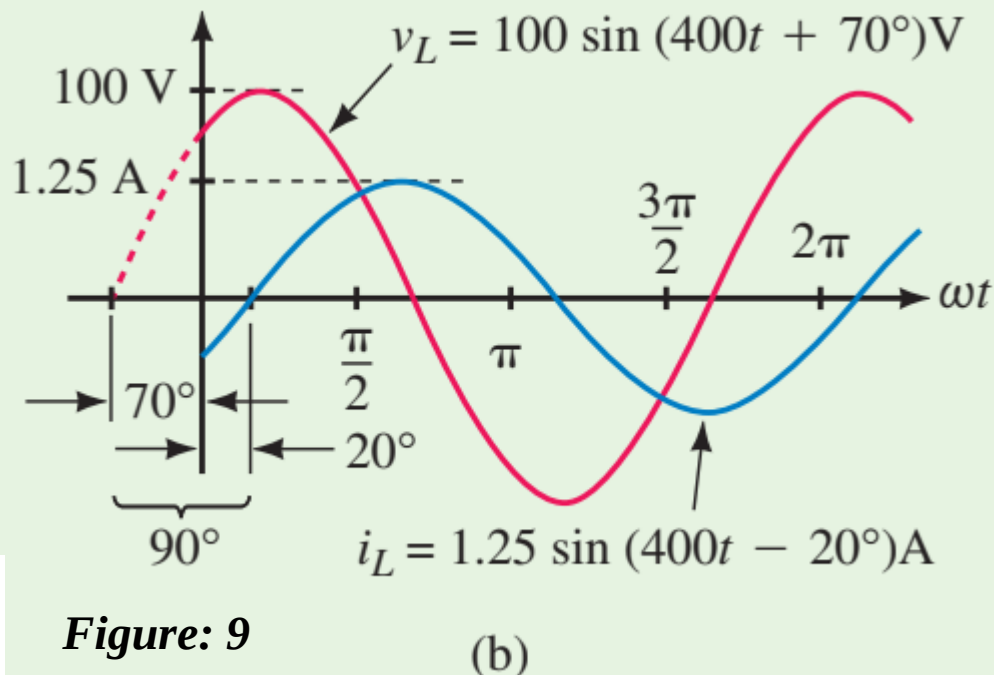
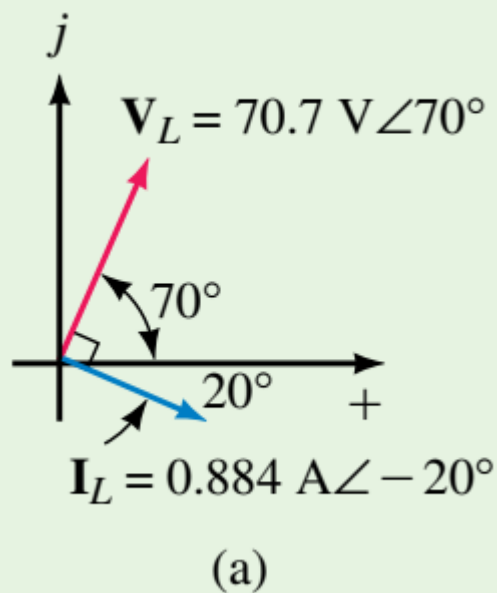
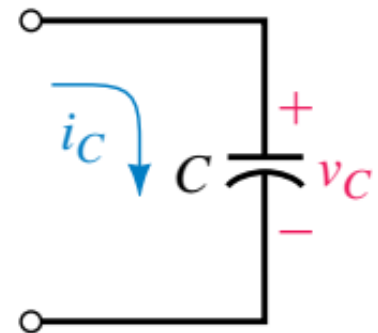


Figure: 9

4.3 Capacitance and Sinusoidal AC

- *Phase Lead in a Capacitive Circuit*
- For capacitance, current is proportional to rate of change of voltage
- If v_c is a sine wave, we get

$$i_C = C \frac{dv_C}{dt} = C \frac{d}{dt}(V_m \sin \omega t) = \omega C V_m \cos \omega t = I_m \cos \omega t$$



$$i_C = C \frac{dv_C}{dt}$$

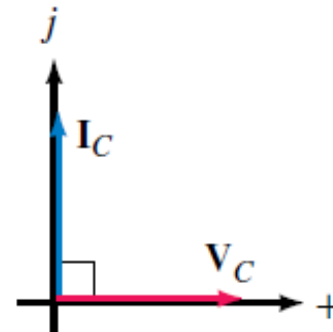
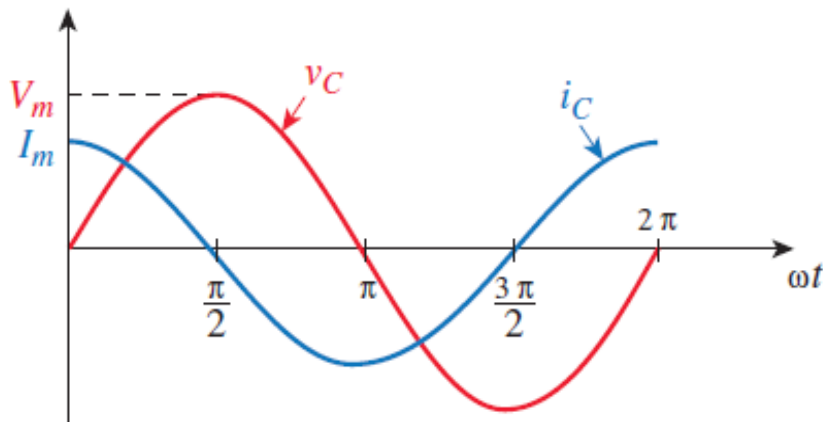
Capacitance and Sinusoidal AC continue...

- Using trigonometric identity $\cos \omega t = \sin(\omega t + 90^\circ)$

$$i_C = I_m \sin(\omega t + 90^\circ)$$

$$I_m = \omega C V_m$$

- For a pure capacitive circuit, current leads voltage by 90°
- Alternatively, voltage lags current by 90°



4.3.1 Capacitive Reactance

- $I_m = \omega C V_m$
- Using above equation the ratio V_m to I_m is:

$$X_C = \frac{V_m}{I_m} \quad (\Omega) \quad \longrightarrow \quad X_C = \frac{1}{\omega C} \quad (\Omega)$$

- This ratio is defined as *capacitive reactance* (X_C).
- Current and voltage amplitudes are related by:

$$I_m = \frac{V_m}{X_C} \quad \& \quad V_m = I_m X_C$$

EXAMPLE 8

The voltage across a 10- μF capacitance is $v_C = 100 \sin(\omega t - 40^\circ)$ V and $f = 1000$ Hz. Determine i_C and sketch its waveform.

Solution

$$\omega = 2\pi f = 2\pi(1000 \text{ Hz}) = 6283 \text{ rad/s}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(6283)(10 \times 10^{-6})} = 15.92 \Omega$$

$$I_m = \frac{V_m}{X_C} = \frac{100 \text{ V}}{15.92 \Omega} = 6.28 \text{ A}$$

EXAMPLE 8

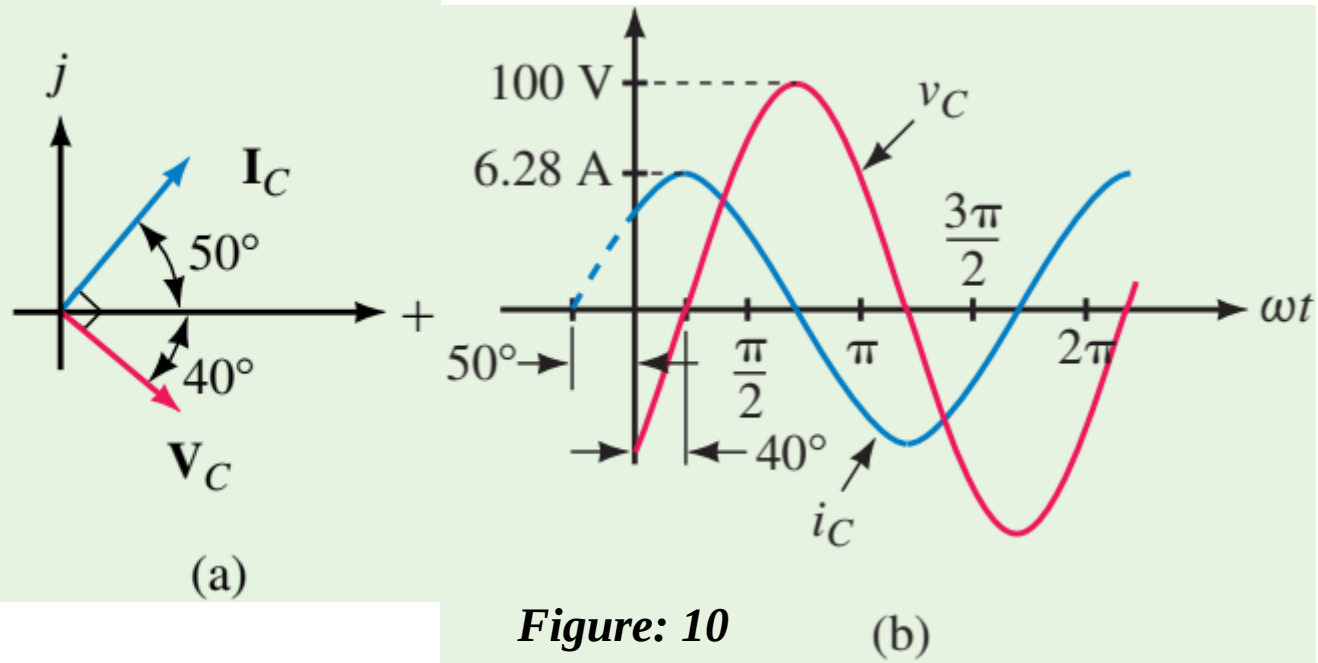


Figure: 10

5. Impedance

- The opposition that a circuit element presents to current in phasor domain (frequency domain) is its **impedance, Z**
 - $Z = V/I \Omega$ (the ratio of voltage phasor to its current phasor)
 - Since phasor voltages and currents are complex, **Z** is also complex

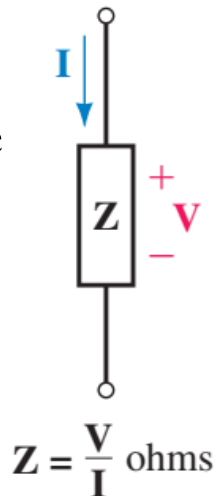
$$Z = \frac{V}{I} \text{ (ohms)}$$

- Where V and I are the rms magnitudes of **V** and **I** respectively and θ is the angle between them, from above equation we get

$$Z = Z \angle \theta$$

- Once the impedance of a circuit is know the current and voltage can be determined using

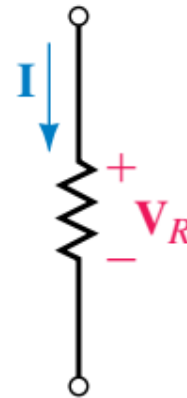
$$I = \frac{V}{Z} \quad \& \quad V = IZ$$



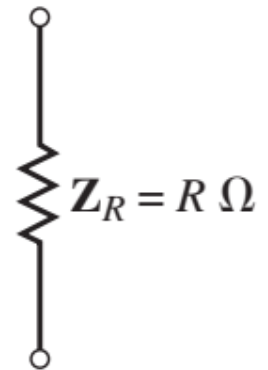
5.1 Resistance

- For a pure resistive circuit, the voltage and current are in phase
- If the voltage has a phase angle, the current has the same angle
- If $\mathbf{V}_R = V_R \angle \theta$ then $\mathbf{I} = I \angle \theta$. Thus impedance

$$\mathbf{Z}_R = \frac{\mathbf{V}_R}{\mathbf{I}} = \frac{V_R \angle \theta}{I \angle \theta} = \frac{V_R}{I} \angle 0^\circ = R \angle 0^\circ = R$$



(a) Voltage and current



(b) Impedance

5.2 Inductance

- For a pure inductive circuit, the current lags voltage by 90°
- Assume a 0° angle for voltage

- $\mathbf{V}_L = V_L \angle 0^\circ$ then $\mathbf{I} = I \angle -90^\circ$. Thus impedance

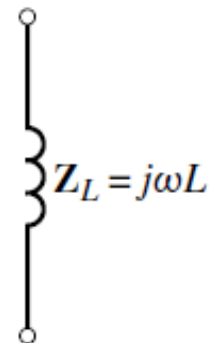
$$\mathbf{Z}_L = \frac{\mathbf{V}_L}{\mathbf{I}} = \frac{V_L \angle 0^\circ}{I \angle -90^\circ} = \frac{V_L}{I} \angle 90^\circ = \omega L \angle 90^\circ = j\omega L$$

- Substitute $V_L / I_L = \omega L$, we get

$$\mathbf{Z}_L = j\omega L = jX_L$$



(a) Voltage and current



(b) Impedance

EXAMPLE 9

Given $v_L = 100 \sin(400t + 70^\circ)$ V and $L = 0.2$ H, determine i_L using the impedance concept.

Solution See Figure 11. Use rms values. Thus,

$$\mathbf{V}_L = 70.7 \text{ V} \angle 70^\circ \quad \text{and} \quad \omega = 400 \text{ rad/s}$$

$$\mathbf{Z}_L = j\omega L = j(400)(0.2) = j80 \, \Omega$$

$$\mathbf{I}_L = \frac{\mathbf{V}_L}{\mathbf{Z}_L} = \frac{70.7 \angle 70^\circ}{j80} = \frac{70.7 \angle 70^\circ}{80 \angle 90^\circ} = 0.884 \text{ A} \angle -20^\circ$$

In the time domain, $i_L = \sqrt{2}(0.884) \sin(400t - 20^\circ) = 1.25 \sin(400t - 20^\circ)$ A.

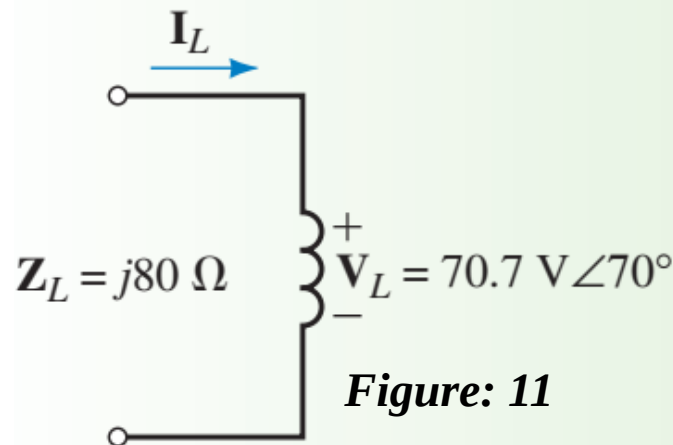


Figure: 11

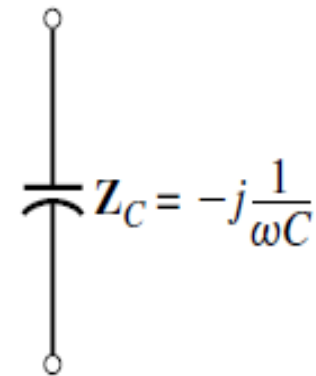
5.3 Capacitance

- For a pure capacitive circuit, the current leads voltage by 90°
- Assume a 0° angle for voltage
 - $\mathbf{V}_c = V_c \angle 0^\circ$ then $\mathbf{I} = I \angle 90^\circ$. Thus impedance

$$\mathbf{Z}_C = \frac{\mathbf{V}_C}{\mathbf{I}} = \frac{V_C \angle 0^\circ}{I \angle 90^\circ} = \frac{V_C}{I} \angle -90^\circ = \frac{1}{\omega C} \angle -90^\circ = -j \frac{1}{\omega C} \quad (\text{ohms})$$

- Thus,

$$\mathbf{Z}_C = -j \frac{1}{\omega C} = -jX_C \quad (\text{ohms})$$



Impedance of a pure capacitance

HOME WORK

EXAMPLE 10

Given $v_C = 100 \sin(\omega t - 40^\circ)$ V, $f = 1000$ Hz, and $C = 10 \mu\text{F}$, determine i_C in Figure 12.

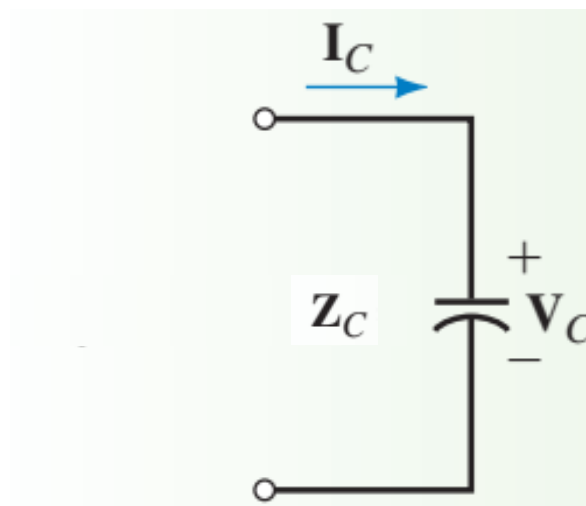


Figure: 12

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*